Biochemistry Cume

Oct. 5, 2001

1. (10 points) Mitochondrial and many bacterial cytochrome c oxidases use cytochrome c as an electron donor for the reduction of oxygen. Write an overall, balanced equation for the process catalyzed by cytochrome c oxidase including any processes which might be coupled to the electron transfer reactions within the complex.

2. (10 points) The structures of bovine and Paracoccus denitrificans cytochrome c oxidase were determined in the mid-1990's. Although greatly different in their size and subunit composition, both were found to have a similar arrangement of redox centers associated with Subunits I and II. Draw a cartoon type of diagram showing the relative positions of these redox centers with respect to the membrane and Subunits I and II.

3. (10 points) In describing the structure function of these and other terminal oxidases the term "binuclear center" is often used. To what does this term refer?

4. (10 points) Analysis of the structures of these cytochrome c oxidases has led to the identification of the "D pathway" and "K pathway". In what processes involving cytochrome c oxidase are these pathways involved.

5. (10 points) In examining the three dimensional structures of these and (the other terminal oxidases covered in the readings), a common secondary structural motif is used to thread the protein back and forth through the membrane. What type of structure is it? Why is this structure particularly good for transmembrane proteins?

6. (10 points) The mitochondrial and Paracoccus denitrificans are often referred to as aa3 oxidases.. In some of the other readings, you encountered the Thermophilus thermophilus ba3 oxidase, the Escherichia coli bo3 oxidase, and the Rhodothermus marinus caa3 oxidase. What are these designations aa3, ba3, bo3, and caa3) meant to indicate?

7. (10 points) The latter three oxidases in question 6 all reduce oxygen to water but have different electron donors. What is thought to be the electron donor to the terminal oxidase in each case? How does the site of entry of electrons into the terminal oxidase differ in these cases.

8. (15 points) Of the latter three oxidases in question 7, which would you suggest to be the best model for mitochondrial cytochrome c oxidase? Write a paragraph defending your choice.

9. (15 points) Of the latter three oxidases in question 7, which would you suggest to be the worst model for mitochondrial cytochrome c oxidase? Write a paragraph defending your choice.